Technical Document #1044: Database Systems - Comprehensive Overview

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Database monitoring tracks performance metrics and identifies bottlenecks. Query analysis and indexing strategies optimize database performance.

Relational databases organize data into tables with rows and columns. SQL is used to query and manipulate structured data.

Sharding distributes data across multiple database servers. It improves performance by parallelizing query processing.

Database replication copies data across multiple servers. It improves availability and provides fault tolerance.

Database indexing improves query performance by creating data structures that allow faster data retrieval.

NoSQL databases handle unstructured and semi-structured data. Document stores, key-value stores, and graph databases are common NoSQL types.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Key Takeaways:

- Database replication copies data across multiple servers.
- Database caching stores frequently accessed data in memory.
- ACID properties (Atomicity, Consistency, Isolation, Durability) ensure reliable transaction processing in databases.